

Администрирование локальных сетей

Лабораторная работа №5

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Цели и задачи работы

Получить основные навыки по настройке виртуальных локальных сетей (VLAN) на коммутаторах Cisco и изучить принципы сегментации сети.

Выполнение лабораторной работы

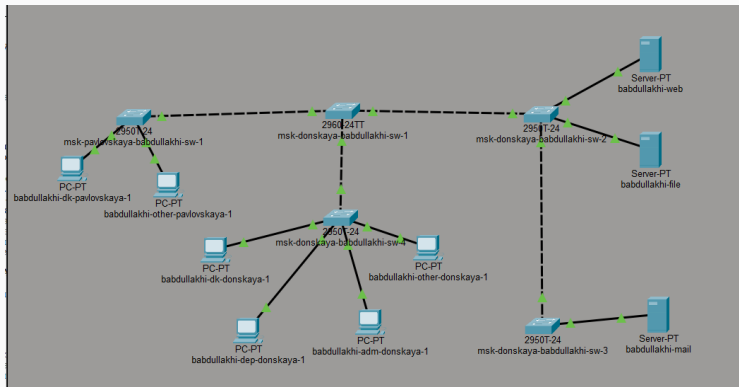


Рис. 1: Топология сети

Настройка VTP-сервера

```
msk-donskaya-babdullakhi-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-babdullakhi-sw-1(config)#vtp domain donskeya
Changing VTP domain name from NULL to donskeya
msk-donskaya-babdullakhi-sw-1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
donskeya.

msk-donskaya-babdullakhi-sw-1(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-babdullakhi-sw-1(config)#vlan 2
msk-donskaya-babdullakhi-sw-1(config-vlan)#name management
msk-donskaya-babdullakhi-sw-1(config-vlan)#vlan 3
msk-donskaya-babdullakhi-sw-1(config-vlan)#name servers
msk-donskaya-babdullakhi-sw-1(config-vlan)#vlan 101
msk-donskaya-babdullakhi-sw-1(config-vlan)#name dk
msk-donskaya-babdullakhi-sw-1(config-vlan)#vlan 102
msk-donskaya-babdullakhi-sw-1(config-vlan)#name departaments
msk-donskaya-babdullakhi-sw-1(config-vlan)#vlan 103
msk-donskaya-babdullakhi-sw-1(config-vlan)#name adm
msk-donskaya-babdullakhi-sw-1(config-vlan)#vlan 104
msk-donskaya-babdullakhi-sw-1(config-vlan)#name other
msk-donskaya-babdullakhi-sw-1(config-vlan)#int f0/24
msk-donskaya-babdullakhi-sw-1(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-1(config-if)#int g0/1
msk-donskaya-babdullakhi-sw-1(config-if)#sw mode trunk

msk-donskaya-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#int g0/2
msk-donskaya-babdullakhi-sw-1(config-if)#sw mode trunk

msk-donskaya-babdullakhi-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-babdullakhi-sw-1(config-if)#int f0/1
msk-donskaya-babdullakhi-sw-1(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-1(config-if)#exit
```

Настройка коммутатора sw-2

```
msk-donskaya-babdullakhi-sw-2>ena
msk-donskaya-babdullakhi-sw-2>enable
Password:
msk-donskaya-babdullakhi-sw-2#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-sw-2(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-babdullakhi-sw-2(config)#vtp domain donsкаya
Domain name already set to donsкаya.
msk-donskaya-babdullakhi-sw-2(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-babdullakhi-sw-2(config)#int g0/1
msk-donskaya-babdullakhi-sw-2(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-2(config-if)#int g0/2
msk-donskaya-babdullakhi-sw-2(config-if)#sw mode trunk

msk-donskaya-babdullakhi-sw-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-babdullakhi-sw-2(config-if)#int range f0/1-f0/2
msk-donskaya-babdullakhi-sw-2(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-2(config-if-range)#sw acc vlan 3
msk-donskaya-babdullakhi-sw-2(config-if-range)#exit
msk-donskaya-babdullakhi-sw-2(config)#exit
msk-donskaya-babdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-2#wr mem
Building configuration...
```

Рис. 3: Настройка коммутатора sw-2

Настройка коммутатора sw-3

```
msk-donskaya-babdullakhi-sw-3>ena
msk-donskaya-babdullakhi-sw-3>enable
Password:
msk-donskaya-babdullakhi-sw-3#vtp mode cli
      ^
% Invalid input detected at '^' marker.

msk-donskaya-babdullakhi-sw-3#
msk-donskaya-babdullakhi-sw-3#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-sw-3(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-donskaya-babdullakhi-sw-3(config)#vtp domain donsokaya
Changing VTP domain name from NULL to donsokaya
msk-donskaya-babdullakhi-sw-3(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
donsokaya.

msk-donskaya-babdullakhi-sw-3(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-babdullakhi-sw-3(config)#int g0/1
msk-donskaya-babdullakhi-sw-3(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-3(config-if)#int ran f0/1-f0/2
msk-donskaya-babdullakhi-sw-3(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-3(config-if-range)#sw acc vlan 3
msk-donskaya-babdullakhi-sw-3(config-if-range)#exit
msk-donskaya-babdullakhi-sw-3(config)#exit
msk-donskaya-babdullakhi-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-3#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-sw-3#
```

Рис. 4: Настройка коммутатора sw-3

Настройка коммутатора sw-4

```
msk-donskaya-babdullakhi-sw-4>enab
Password:
msk-donskaya-babdullakhi-sw-4#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-sw-4(config)#vtp mode cli
Device mode already VTP CLIENT.
msk-donskaya-babdullakhi-sw-4(config)#vtp domain donsкаya
Domain name already set to donsкаya.
msk-donskaya-babdullakhi-sw-4(config)#vtp pass cisco
Password already set to cisco
msk-donskaya-babdullakhi-sw-4(config)#int g0/1
msk-donskaya-babdullakhi-sw-4(config-if)#sw mode trunk
msk-donskaya-babdullakhi-sw-4(config-if)#int r f0/1-f0/5
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw acc vlan 101
msk-donskaya-babdullakhi-sw-4(config-if-range)#int r f0/6-f0/10
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw acc vlan 102
msk-donskaya-babdullakhi-sw-4(config-if-range)#int r f0/11-f0/15
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw acc vlan 103
msk-donskaya-babdullakhi-sw-4(config-if-range)#int r f0/16-f0/20
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw mode acc
msk-donskaya-babdullakhi-sw-4(config-if-range)#sw acc vlan 104
msk-donskaya-babdullakhi-sw-4(config-if-range)#exit
msk-donskaya-babdullakhi-sw-4(config)#exit
msk-donskaya-babdullakhi-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-sw-4#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-sw-4#
```

Настройка коммутатора pavlovskaya-sw-1

```
msk-pavlovskaya-babdullakhi-sw-1>enab
Password:
msk-pavlovskaya-babdullakhi-sw-1#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
msk-pavlovskaya-babdullakhi-sw-1(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-pavlovskaya-babdullakhi-sw-1(config)#vtp domain donskaya
Domain name already set to donskaya.
msk-pavlovskaya-babdullakhi-sw-1(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-pavlovskaya-babdullakhi-sw-1(config)#int f0/24
msk-pavlovskaya-babdullakhi-sw-1(config-if)#sw mode trunk
msk-pavlovskaya-babdullakhi-sw-1(config-if)#int r f0/1/f0/15
      ^
% Invalid input detected at '^' marker.

msk-pavlovskaya-babdullakhi-sw-1(config-if)#
msk-pavlovskaya-babdullakhi-sw-1(config-if)#int r f0/1-f0/15
msk-pavlovskaya-babdullakhi-sw-1(config-if-range)#sw mode acc
msk-pavlovskaya-babdullakhi-sw-1(config-if-range)#sw acc vlan 101
msk-pavlovskaya-babdullakhi-sw-1(config-if-range)#int f0/20
msk-pavlovskaya-babdullakhi-sw-1(config-if)#sw mode acc
msk-pavlovskaya-babdullakhi-sw-1(config-if)#sw acc vlan 104
msk-pavlovskaya-babdullakhi-sw-1(config-if)#exit
msk-pavlovskaya-babdullakhi-sw-1(config)#exit
msk-pavlovskaya-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
msk-pavlovskaya-babdullakhi-sw-1#
```

Настройка IP-адресов серверов

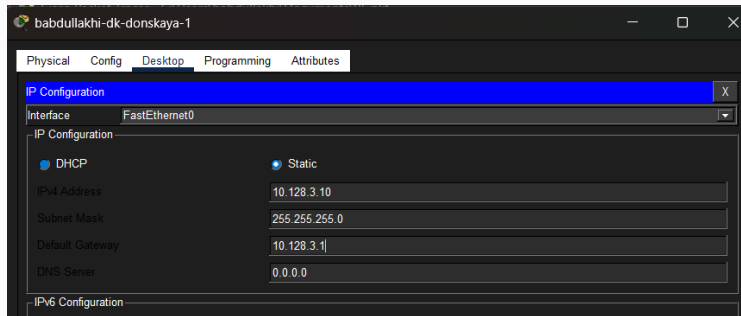


Рис. 7: Настройка IP серверов

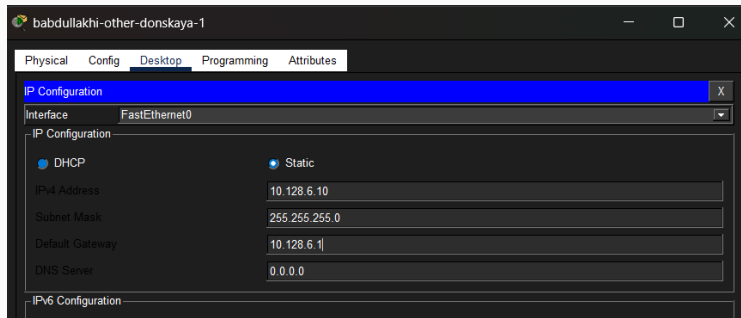
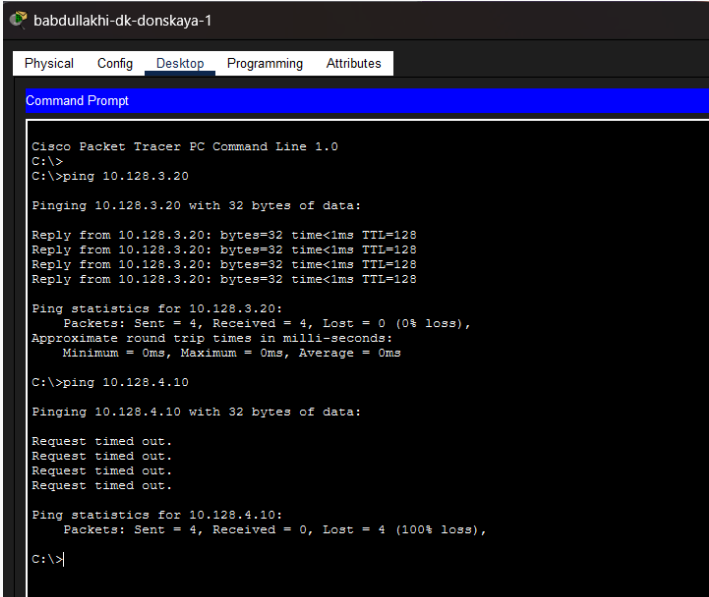


Рис. 8: Настройка IP ПК

Проверка соединения



The screenshot shows the Cisco Packet Tracer Desktop environment. At the top, there are tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is selected, displaying a Command Prompt window titled "Command Prompt". The window shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.128.3.20

Pinging 10.128.3.20 with 32 bytes of data:

Reply from 10.128.3.20: bytes=32 time<1ms TTL=128
Reply from 10.128.3.20: bytes=32 time<1ms TTL=128
Reply from 10.128.3.20: bytes=32 time<1ms TTL=128
Reply from 10.128.3.20: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.10

Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

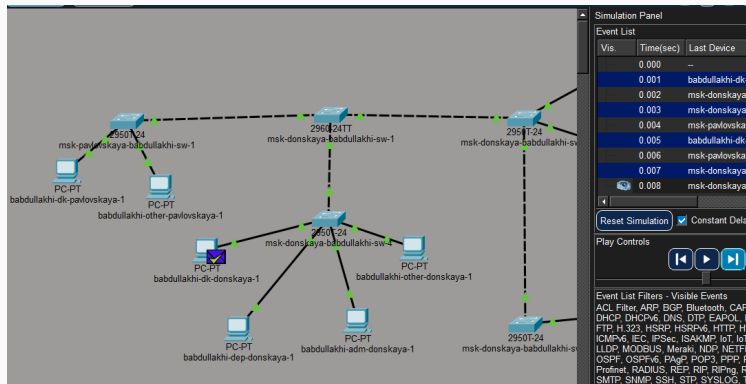


Рис. 10: Режим Simulation

Структура ICMP-пакета

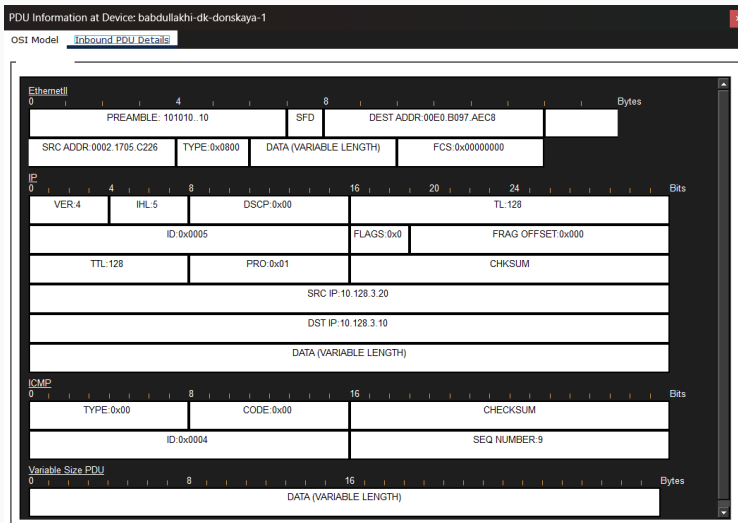


Рис. 11: Структура ICMP-пакета

Вывод

В ходе лабораторной работы была построена сеть в Cisco Packet Tracer и выполнена настройка VLAN на коммутаторах. Один из коммутаторов был настроен как VTP-сервер, а остальные — как VTP-клиенты.

Были настроены trunk-соединения между коммутаторами и выполнено распределение портов по VLAN. Также на серверах и рабочих станциях были назначены статические IP-адреса.

Проверка соединений показала корректную работу сети: устройства внутри одной VLAN успешно обмениваются данными, а устройства разных VLAN изолированы. Дополнительно был изучен процесс передачи ICMP-пакетов в режиме Simulation.